



Trajectory tracking control of a 6-DOF robotic arm based on improved FOPID

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Abstract

Robotic arm is a complex system with multiple inputs and outputs, strong nonlinearity and strong coupling, and the research of high precision trajectory tracking control technology for robotic arm has been an important issue for scholars at home and abroad. This paper takes the six-degree-of-freedom (6-DOF) robotic arm as its study object and designs a fractional-order PID (FOPID) control method. To improve its control accuracy, a parameter tuning method of fractional-order beetle antennae particle swarm algorithm (FBPA) optimized FOPID controller is proposed. This method puts the beetle antennae search (BAS) algorithm together with the particle swarm optimization (PSO) algorithm, introduces the concept of fractional-order calculus into the algorithm, dynamically adjusts the inertial weights and fractional order and finally improves the optimization effect of the algorithm. The simulation experiments of MATLAB/Simulink indicate that in comparison with the traditional PID control method, the FOPID control method optimized by the FBPA has high control accuracy and small overshooting, which meets the high-precision control requirements of the 6-DOF robotic arm.

Keywords 6-DOF robotic arm · Trajectory tracking · Optimization algorithm · FOPID controller

1 Introduction

In the context of the new era of industrial revolution, the application of robotic arms has been increasing significantly, due to their high intelligence, ability to perceive the external environment and repeatable programming. Therefore, their demand in medical care, military, industrial manufacturing, etc., is increasing year by year [1, 2]. However, as a complex mechanism system with multiple inputs and outputs, strong coupling and strong nonlinearity, the robotic arm often has uncertain factors during movement such as external interference and measurement errors. Its end position sometimes completely deviates from the preset reference trajectory, and

thus, one difficult point to study robot arm is the trajectory tracking control method.

When it comes to the domain of industrial robots, PID control is a model-independent control method with a simple controller structure and wide application [3]. However, it has problems such as static error, slow response time and unstable output in tracking control. In addition, PID controllers rely on the numerical model of the controlled system, but sometimes the system is highly nonlinear or has time-varying uncertainties, the parameter tuning process is complex, and thus, it is difficult to accurately model and has poor adaptability to different working conditions. When there is noise or external interference in the control system, poor tuning and poor performance may occur.

In order to break the limitations of PID controllers, the academic community has widely adopted computational intelligence technologies such as genetic algorithms, fuzzy logic control and neural networks to control robots and complex nonlinear systems. Hybrid methods to combine PID controllers with other control methods have also been adopted [4–6]. Recently, fractional-order calculus theory has gradually been applied to controller design research. FOPID controllers not only have some characteristics of traditional integer-order PID controllers, but also have certain advan-

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tages in control accuracy and anti-interference ability. This also provides a new idea for the selection and implementation of robotic arm trajectory tracking.

In comparison with the traditional PID controller, the FOPID controller increases the selection of integral-order λ and differential-order μ and has better robustness and control effect, but at the same time, it makes parameter adjustment more complicated. How to apply the FOPID controller to the field of industrial control and perform parameter optimization has become a research hotspot. At present, scholars at home and abroad are committed to the study of parameter tuning of FOPID controllers using swarm intelligence optimization algorithms. Reference [7] designed a FOPID controller based on the gray wolf optimization algorithm and applied it to the DFIG wind farm and achieved good control effect. Reference [8] uses the whale optimization algorithm to optimize the parameters of the FOPID controller and applies it to load frequency control of the power system. The performance of the whale algorithm is compared with other optimization methods through simulation experiments. According to the results, the FOPID controller optimized by the whale algorithm has the advantages of small overshoot and short adjustment time.

The 6-DOF robotic arm is composed of six connecting rods in series, which is suitable for working at any angle or trajectory. Due to the rapid development of computer technology, the control of 6-DOF robotic arm needs more and higher requirements. Today's control methods are combined with advanced control algorithms to better handle the 6-DOF robotic arm's own errors and external disturbances. Reference [9] used a new vision-based calibration framework to maintain the accuracy of a 6-DOF robotic arm and achieved good results. Reference [10] used the frequency response function to improve the cutting stability of a 6-DOF machining robot. Reference [11] solved the movement problem of a 6-DOF robotic arm in complex scenes by applying a deep reinforcement learning method. Reference [12] used grey wolf optimization algorithm to implement the optimal trajectory generation method for a 6-DOF parallel robotic arm. Reference [13] applied grey wolf algorithm to the inverse kinematics analysis of a 6-DOF oil and gas pipeline welding robotic arm and achieved good results.

This paper mainly studies the design and application of FOPID based on FBPA, and the main contributions are as follows: Firstly, the BAS algorithm is introduced based on the PSO algorithm, with fractional calculus incorporated into the velocity iteration update formula, leading to the proposed FBPA. The results indicate that in comparison with the traditional optimization methods, the FBPA improves the convergence speed and accuracy to a great extent. Secondly, in view of the complexity and uncertainty of FOPID parameter tuning, the FBPA is used to optimize and tune FOPID parameters. Finally, trajectory tracking experiments are car-

ried out to verify the effectiveness of the FBPA and to prove that FOPID has higher control accuracy with a 6-DOF robotic arm.

2 Preliminaries

This section introduces the definition of fractional-order calculus, the definitions related to FOPID and the mathematical model used for establishing a 6-DOF robotic arm.

2.1 Fractional-order calculus

A uniform operator of fractional-order calculus will be introduced here, where α is the order of integration or differentiation, t is the independent variable, and t_0 is the lower bound of this variable. The specific expression form is as follows:

$${}_t D_t^\alpha f(t) = \begin{cases} \frac{d^\alpha}{dt^\alpha} f(t), & \alpha > 0 \\ f(t), & \alpha = 0 \\ \int_{t_0}^t f(\tau) d\tau^{-\alpha}, & \alpha < 0 \end{cases} \quad (1)$$

where ${}_t D_t^\alpha f(t)$ denotes the α -order derivative of the function concerning the independent variable t if $\alpha > 0$; $\alpha = 0$ denotes the original signal; and $-\alpha$ denotes the order integral if $\alpha < 0$.

The fractional-order calculus has many definitions: starting from the n-fold definition to other different variants related to the definition. The three most commonly used definitions in control systems are: the Grünwald–Letnikov (G-L) definition, the Riemann–Liouville (R-L) definition and the Caputo definition [14].

The Grünwald–Letnikov (G-L) definition:

$${}_t D_t^\alpha = \lim_{h \rightarrow 0} \frac{1}{h^\alpha} \sum_{j=0}^{\left[\frac{t-t_0}{h}\right]} (-1)^j \binom{\alpha}{j} f(t-jh). \quad (2)$$

where $\left[\frac{t-t_0}{h}\right]$ is the integer part, and t_0 and t are the lower and upper limits of the operator, respectively. The values of the binomial coefficients are given by:

$$\binom{\alpha}{j} = \frac{\Gamma(\alpha+1)}{\Gamma(j+1) \cdot \Gamma(\alpha-j+1)}. \quad (3)$$

The gamma function in the above equation can be defined by the following equation:

$$\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt, \quad x > 0. \quad (4)$$

The Riemann–Liouville (R-L) definition:

$${}_t D_t^\alpha f(t) = \frac{1}{\Gamma(m-\alpha)} \cdot \frac{d^m}{dt^m} \int_{t_0}^t \frac{f(\tau)}{(t-\tau)^{1+\alpha-m}} d\tau. \quad (5)$$

where t and t_0 refer to the upper and lower limits of the operator, respectively; m is an integer satisfying $m-1 < \alpha \leq m$.

The Caputo definition:

$${}_t D_t^\alpha f(t) = \frac{1}{\Gamma(n-\alpha)} \cdot \frac{d^n}{dt^n} \int_{t_0}^t \frac{f^n(\tau)}{(t-\tau)^{1+\alpha-n}} d\tau. \quad (6)$$

where t and t_0 refer to the upper and lower limits of the operator, respectively; n is an integer satisfying $n-1 < \alpha \leq n$.

2.2 PID controller and FOPID controller

Based on the feedback concept in traditional control theory, a PID controller is designed to achieve effective control of the system by adjusting three key parameters. The purpose of the controller is to ensure that the output signal can accurately follow the input signal [15]. PID controller's structure is shown in Fig. 1:

The input of the PID controller is the error $e(t)$, which is defined as follows:

$$e(t) = r(t) - y(t). \quad (7)$$

where $r(t)$ is the desired value, $y(t)$ is the system output value, and $e(t)$ is the error value. The transfer function of the PID control system can be expressed as:

$$G_C(s) = \frac{U(s)}{E(s)} = K_p + \frac{K_i}{s} + K_d s. \quad (8)$$

where K_p , K_i , K_d are the proportional coefficient, integral coefficient and differential coefficient, respectively.

The transfer function of the FOPID controller is:

$$G_C(s) = K_p + \frac{K_i}{s^\lambda} + K_d s^\mu. \quad (9)$$

where λ , μ are all greater than 0.

Compared with the traditional PID controller, there are two more adjustable parameters for the FOPID controller, the integral-order λ and the differential-order μ , and therefore have better control effect [16–18]. In addition, it shows stronger adaptability, can reflect the characteristics of non-linear dynamic systems more accurately and is suitable for various types of systems. However, the additional parameters will also make its parameter setting more complicated,

so the intelligent optimization algorithm is adopted to realize the parameter setting of the FOPID controller. Figure 2 illustrates FOPID controller's structure diagram.

2.3 Establishing the dynamic model of the 6-DOF robotic arm

The purpose of this paper is to design an improved FOPID controller to achieve high-performance tracking control for a 6-DOF robotic arm. The UR5 robotic arm, a 6-DOF industrial robot, is used as the modeling object, as shown in Fig. 3. In this section, the Lagrange modeling method is employed to establish the dynamic model of the UR5 robotic arm. This model is then used as the controlled object in simulations to verify the effectiveness of the proposed FOPID controller.

The Lagrange function of the robotic arm system is defined as follows [19]:

$$L = K - P. \quad (10)$$

where L represents the Lagrange function, K is the system kinetic energy, and P is the system potential energy. The model equation of the system dynamics state is as follows:

$$\tau_i = \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i}, i = 1, 2, \dots, n. \quad (11)$$

where τ_i is the generalized force or generalized torque acting on the i -th generalized coordinate, q_i refers to the angle of the i -th joint, and $\dot{q}_i$ is the corresponding angular velocity of the i -th joint.

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}} - \frac{\partial K}{\partial q} + \frac{\partial P}{\partial q} = \tau. \quad (12)$$

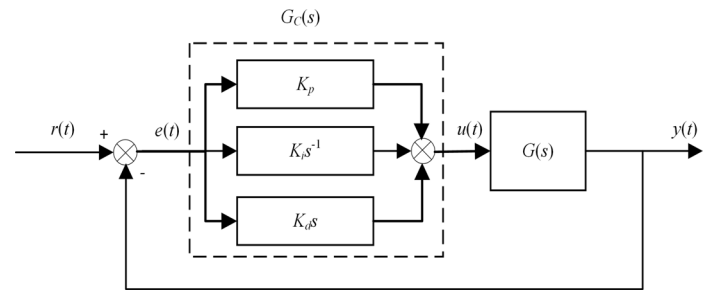
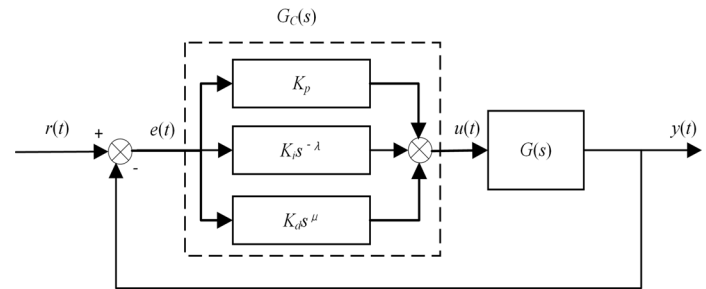
(1) Calculate the total kinetic energy K . Set the velocity of any point on connecting rod i as follows:

$$v = \sum_{j=1}^i \left(\frac{\partial T_i}{\partial \dot{q}_j} \dot{q}_j \right)^i r. \quad (13)$$

The position vector of point i r on link i is given, where the mass is dm . The kinetic energy of link i at this point is calculated as:

$$dK_i = \frac{1}{2} v^2 dm = \frac{1}{2} \text{tr} \left[\sum_{j=1}^i \sum_{k=1}^i \frac{\partial T_i}{\partial \dot{q}_j} \left({}^i r dm {}^i r^T \right) \frac{\partial T_i}{\partial \dot{q}_k} \dot{q}_j \dot{q}_k \right] dm. \quad (14)$$

By summing the kinetic energy of n connecting rods, we can know the kinetic energy of the system at a certain rod;

Fig. 1 The structure of PID controller**Fig. 2** The structure of FOPID controller

that is, the total kinetic energy of the robotic arm system is:

$$K = \int dK_i = \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^i \sum_{k=1}^i \text{tr} \left(\frac{\partial_i^0 T}{\partial q_j} I_i \frac{\partial_i^0 T^T}{\partial q_k} \right) \dot{q}_j \dot{q}_k. \quad (15)$$

where I_i represents the pseudo-inertia matrix.

- (2) Calculate the total potential energy P . Define the position vector ${}^i r$ on the connecting rod i , whose mass is dm and whose potential energy at this point is:

$$dP_i = -dm g^T r = -g^T T_i^T {}^i r dm. \quad (16)$$

$$\text{where } g^T = [g_x \ g_y \ g_z \ 1].$$

The total potential energy of a robotic arm with n links is the sum of the potential energies of each link:

$$P = \sum_{i=1}^n P_i = - \sum_{i=1}^n m_i g^T T_i^T {}^i r_i. \quad (17)$$

Substituting the total kinetic energy and total potential energy formula of the n -DOF manipulator into the formula, the Lagrange dynamic equation is obtained:

$$L = K - P$$

Fig. 3 The UR5 Robotic Arm

$$= \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^i \sum_{k=1}^i \text{tr} \left(\frac{\partial T_i}{\partial q_j} I_i \frac{\partial T_i^T}{\partial q_k} \right) \dot{q}_j \dot{q}_k + \sum_{i=1}^n m_i g^T T_i^T {}^i r_i. \quad (18)$$

The torque at the revolute joint of link i is:

$$\tau_i = \sum_{j=1}^n D_{ij} \ddot{q}_j + I_i \ddot{q}_i + \sum_{j=1}^n \sum_{k=1}^n D_{ijk} \dot{q}_j \dot{q}_k + D_i = M(q) \ddot{q} + C(q, \dot{q}) \dot{q} + G(q). \quad (19)$$

where

$$D_{ij} = \sum_{p=\max(i,j)}^n \text{tr} \left(\frac{\partial T_p}{\partial q_j} I_p \frac{\partial T_p^T}{\partial q_i} \right) \\ D_{ijk} = \sum_{p=\max(i,j,k)}^n \text{tr} \left(\frac{\partial^2 T_p}{\partial q_j \partial q_k} I_p \frac{\partial T_p^T}{\partial q_i} \right) \\ D_i = \sum_{p=i}^n -m_p g^T \frac{\partial T_p}{\partial q_i} p^{r_p} \quad (20)$$

where $q, \dot{q}, \ddot{q} \in \mathbb{R}^n$ are the manipulator joint angle vector, angular velocity vector, and angular acceleration vector, respectively; $M(q) = D_{ij} \in \mathbb{R}^{n \times n}$ is the positive definite

inertia matrix, $C(q, \dot{q}) = D_{ijk} \in R^{n \times n}$ refers to the centripetal force and Coriolis force matrix, and $G(q) = D_i \in R^n$ refers to the gravity matrix. $\tau \in R^n$ stands for the control input matrix of the manipulator system.

In addition, considering the friction between the joints of the robotic arm system, the final dynamics model of the 6-DOF robotic arm is expressed as

$$M(q) + C(q, \dot{q}) + G(q) + \tau_f(\dot{q}) = \tau \quad (21)$$

where $\tau_f(\dot{q}) = [\tau_{f1}(\dot{q}_1), \dots, \tau_{fi}(\dot{q}_i)]$, $i = 1, \dots, n$ represents the friction vector, and $\tau_{fi}(\dot{q}_i) = f_{ci} \text{sgn}(\dot{q}_i) + b_i \dot{q}_i$ with f_{ci} and b_i being friction parameters.

3 Fractional-order beetle antennae particle swarm algorithm

3.1 Improved PSO algorithm

The PSO algorithm is one of the most famous group-based optimization algorithms. It simulates the social behavior of bird flocks to get food. Combining individual and social experience, this group of birds approach their food, continuously change their position according to their own optimal position and the entire group's optimal position and finally reform the optimal queue [20, 21].

In the PSO algorithm, the updating formulas of speed and position are as follows:

$$\begin{aligned} v_{id}^{k+1} &= v_{id}^k + c_1 r_1(t) [p_{id}(t) - x_{id}(t)] \\ &+ c_2 r_2(t) [p_{gd}(t) - x_{id}(t)] \\ x_{id}^{k+1} &= x_{id}^k + v_{id}^{k+1}. \end{aligned} \quad (22)$$

where k refers to the number of iterations, v_{id}^k and x_{id}^k are the velocity and position of the i -th particle in the d -th dimension, c_1 and c_2 refer to learning factors, r_1 and r_2 refer to two random numbers uniformly distributed between $[0, 1]$, and $p_{id}(t)$ and $p_{gd}(t)$ stand for the best historical positions of individual and group particles.

Because the particles run at a high speed, the traditional PSO algorithm is often prone to premature convergence. Since its birth in 1995, researchers have attempted to improve the traditional PSO algorithm in various ways and derive new versions. In order to shun the above-mentioned problem of premature convergence, the concept of inertia weight is introduced into the algorithm. The inertia weight formula is as follows:

$$w = \frac{(w_{\max} - w_{\min}) * k}{\text{MaxIter}}. \quad (23)$$

where w is the inertia weight, $w_{\min}$ refers to the minimum value of the inertia weight set, $w_{\max}$ refers to the maximum value of the inertia weight set, k stands for the current number of iterations, and MaxIter stands for the maximum number of iterations of the algorithm.

Introducing the concept of inertia weight into the velocity update formula, we get an improved velocity update formula:

$$\begin{aligned} v_{id}^{k+1} &= w v_{id}^k + c_1 r_1(t) [p_{id}(t) - x_{id}(t)] \\ &+ c_2 r_2(t) [p_{gd}(t) - x_{id}(t)]. \end{aligned} \quad (24)$$

By introducing inertia weights, particles maintain strong global search capabilities in the early stages of iteration and help to conduct more accurate local searches in the later stages of iteration, reducing particle oscillations and the probability of the population falling into pseudo-extremes [22, 23]. Besides, this paper uses the long memory characteristics of fractional calculus and combines the concept of fractional calculus with the PSO algorithm to describe the search and optimization process of the particle swarm [24, 25]. Fractional calculus is introduced into the speed iteration update formula, and the speed state information of the particles is changed through the historical trajectory information of each particle. Finally, the speed iteration update formula of the fractional PSO algorithm using linearly decreasing inertia weights is obtained:

$$\begin{aligned} v_{id}^{k+1} &= (w - 1 + a) v_{id}^k + \frac{1}{2} a (1 - a) v_{id}^{k-1} \\ &+ \frac{1}{6} a (1 - a) (2 - a) v_{id}^{k-2} \\ &+ \frac{1}{24} a (1 - a) (2 - a) (3 - a) v_{id}^{k-3} \\ &+ c_1 r_1(t) [p_{id}(t) - x_{id}(t)] \\ &+ c_2 r_2(t) [p_{gd}(t) - x_{id}(t)]. \end{aligned} \quad (25)$$

3.2 Improved BAS algorithm

On the basis of the function of the beetle antennae and their random walk mechanism, the BAS algorithm is proposed. According to biological research, beetles locate its food by the concentration of food odor; that is, they find the global optimal value based on the concentration difference of food odor perceived by their two antennae [26–28]. The advantage of the BAS algorithm is that the number of populations is only 1 during the search process, which greatly reduces the amount of calculation of the algorithm. The disadvantage is that it falls into the local optimal situation easily [29, 30].

In order to enhance the algorithm's optimization ability, this study puts the BAS algorithm together with the PSO algorithm and proposes a beetle antennae particle swarm hybrid algorithm. In the beetle antennae particle swarm algorithm, beetles are used to replace particles; likewise, the particle

swarm takes the place of beetle swarm. The beetles update their positions according to two extreme values of fitness values: the optimal values searched by the beetles themselves and the entire beetle swarm separately. In each iteration, the entire beetle population moves towards the optimal solution of the problem.

The speed update formula of each beetle in the beetle antennae particle swarm algorithm realizes the fusion of the speed update formula of BAS and PSO. It effectively overcomes the two weaknesses of PSO algorithm that it is easy to fall into local optimality and low algorithm accuracy. Meanwhile, it enables the BAS algorithm not be restricted to single target parameterization.

$$\begin{aligned}
 v_{id}^{k+1} = & (w - 1 + a)v_{id}^k + \frac{1}{2}a(1 - a)v_{id}^{k-1} \\
 & + \frac{1}{6}a(1 - a)(2 - a)v_{id}^{k-2} \\
 & + \frac{1}{24}a(1 - a)(2 - a)(3 - a)v_{id}^{k-3} \\
 & + c_1r_1(t)[p_{ij}(t) - x_{ij}(t)] \\
 & + c_2r_2(t)[p_{gj}(t) - x_{ij}(t)] + c_3r_3(t)v_i^f.
 \end{aligned} \quad (26)$$

where c_3 has the same value as c_1 and c_2 , which reflects the influence of the search speed of the beetle on the search speed of the particle.

Based on the beetle antennae particle swarm algorithm, the fractional-order α is introduced to make the beetle motion have the inherent properties of fractional calculus. The velocity state information of the beetle is changed by the historical trajectory information of the fractional calculus, which can effectively control the velocity state of the beetle motion and increase the algorithm's optimization ability. The value of α is very important. When α is larger, the beetle's search performance is better, but the search time will be longer. When α is smaller, the beetle will ignore the search results of the previous iteration, and the entire algorithm is prone to be trapped into local optimum. The adjustment formula of the fractional-order α is:

$$\alpha = 0.9 - \frac{0.5 * k}{MaxIter}. \quad (27)$$

To evaluate the optimization performance of the FBPA, four test functions are selected for comparison with the traditional PSO and the improved PSO algorithms. The selected test functions are listed in Table 1, and their corresponding graphs are shown in Fig. 4.

Table 1 Test function

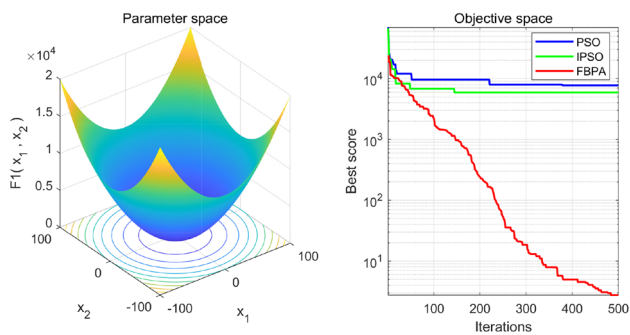
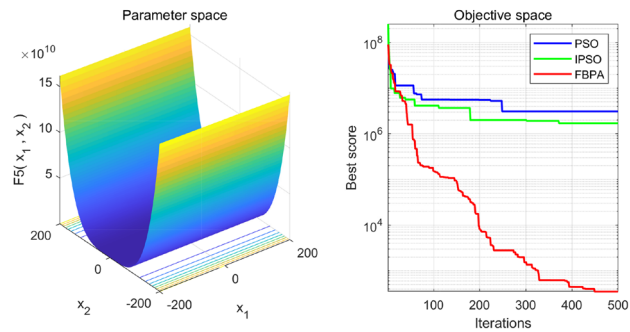
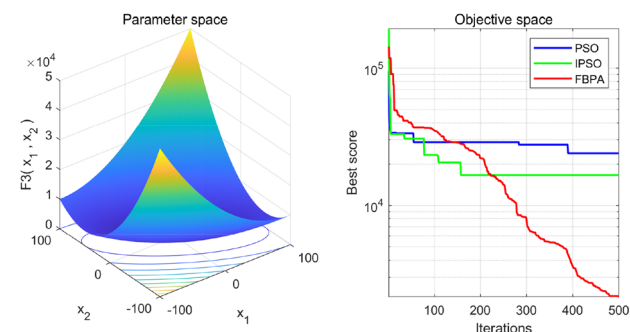
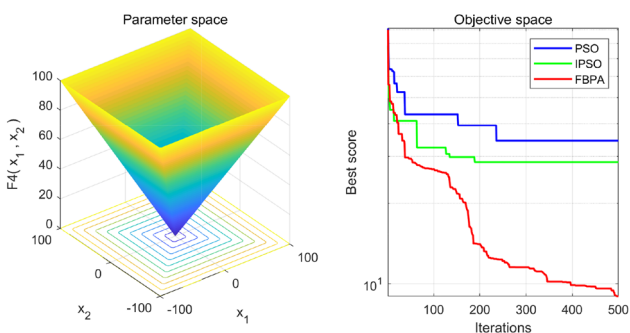
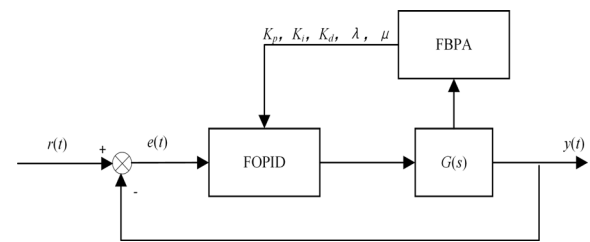
$f(x)$	Dimensions	Range	Minimum
$f_1(x)$	30	$[-100, 100]$	0
$f_2(x)$	30	$[-10, 10]$	0
$f_3(x)$	30	$[-100, 100]$	0
$f_4(x)$	30	$[-100, 100]$	0

The four functions above represent:

$$\begin{aligned}
 f_1(x) &= \sum_{i=1}^n x_i^2 \\
 f_2(x) &= \sum_i^n \left[100 \left(x_{i+1} - x_i^2 \right)^2 + (x_i - 1)^2 \right] \\
 f_3(x) &= \sum_{i=1}^n \left(\sum_{j=1}^i x_j \right)^2 \\
 f_4(x) &= \max_i \{|x_i|, 1 \leq i \leq 30\}
 \end{aligned} \quad (28)$$

According to the optimization results of the benchmark function using the above algorithm, the traditional PSO algorithm encounters local optimal problems too early and cannot avoid being trapped in local optima during the middle and late iterations. The improved PSO achieves a slight enhancement in optimization accuracy and efficiency by dynamically adjusting the inertia weight. However, the proposed FBPA demonstrates the ability to escape local optima, achieving higher optimization accuracy and efficiency compared to both the traditional PSO and the improved PSO algorithms. The control framework is shown in Fig. 5.

Secondly, based on the advantages of the FOPID controller discussed in Sect. 2.2, this paper uses six FOPID controllers to perform trajectory tracking control for the six joints of the 6-DOF robotic arm. The inputs to each FOPID controller are the tracking error, the integral of the tracking error, and the derivative of the tracking error, while the output is the joint torque. Additionally, the parameters to be optimized for each FOPID controller include K_p , K_i , K_d , λ , and μ . To optimize these parameters, this paper employs the FBPA, transforming the parameter optimization problem of the FOPID controller into a swarm optimization problem, where the optimization adjustment of the FOPID controller becomes a task of selecting the global optimal value through the fitness function. Therefore, for the 6-DOF robotic arm system, this paper uses FBPA to optimize the parameters of the six FOPID controllers for the six joints. A total of 30 parameters need to be optimized, specifically K_{pj} , K_{ij} , K_{dj} , λ_j , and μ_j , where $j = 1, 2, 3, 4, 5, 6$ denotes the j -th joint's FOPID controller. Regarding the fitness function, the goal is to achieve the optimal overall tracking error perfor-

(a) $f_1(x)$.(b) $f_2(x)$.(c) $f_3(x)$.(d) $f_4(x)$.**Fig. 4** Optimal convergence graph of $f(x)$ **Fig. 5** The structure diagram of the FOPID controller of the FBPA**Table 2** The physical parameters of the UR5 robotic arm

Joint	Mass (kg)	Length (m)	Inertia matrix (kg · m ²)
Link 1	2.0	0.128	diag[1.0, 1.0, 1.0] × 10 ⁻³
Link 2	2.5	0.612	diag[4.0, 4.0, 4.0] × 10 ⁻²
Link 3	5.7	0.571	diag[6.0, 6.0, 4.0] × 10 ⁻²
Link 4	3.9	0.164	diag[5.5, 5.5, 4.0] × 10 ⁻²
Link 5	2.5	0.115	diag[4.0, 4.0, 4.0] × 10 ⁻²
Link 6	2.5	0.092	diag[4.0, 4.0, 4.0] × 10 ⁻²

mance for the 6-DOF robotic arm system. Therefore, ITAE is selected as the fitness function, expressed as:

$$ITAE = \int_0^{\infty} t \sum_{j=1}^6 |e_j(t)| dt, \quad j = 1, 2, 3, 4, 5, 6. \quad (29)$$

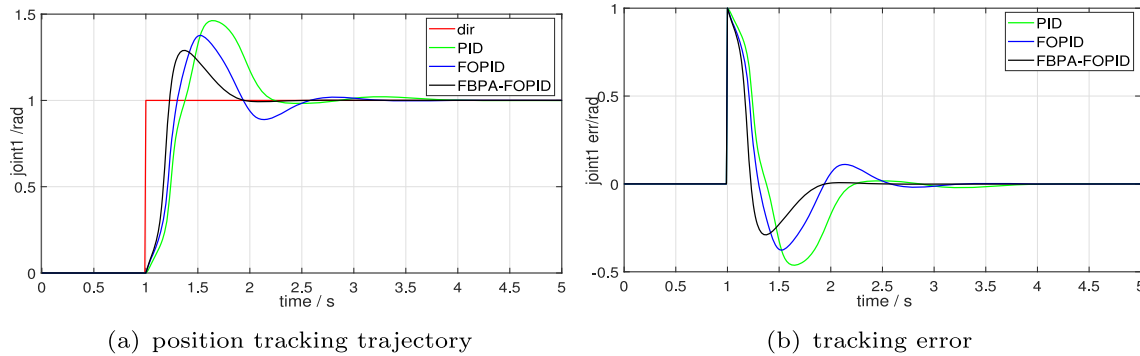
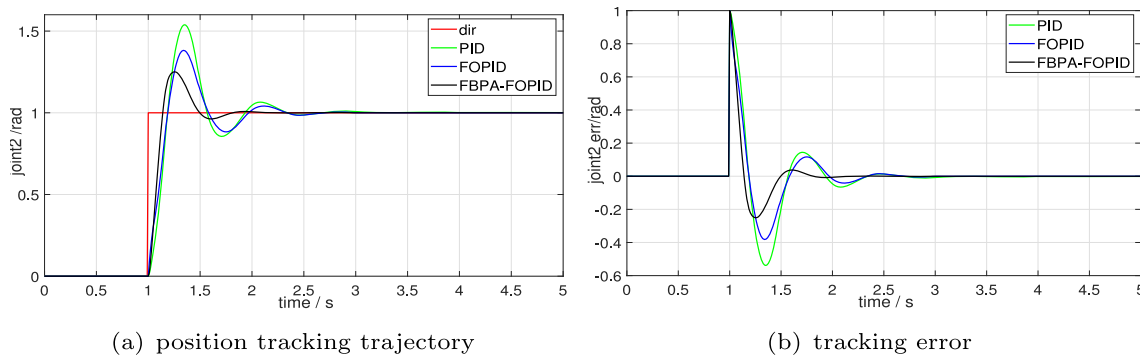
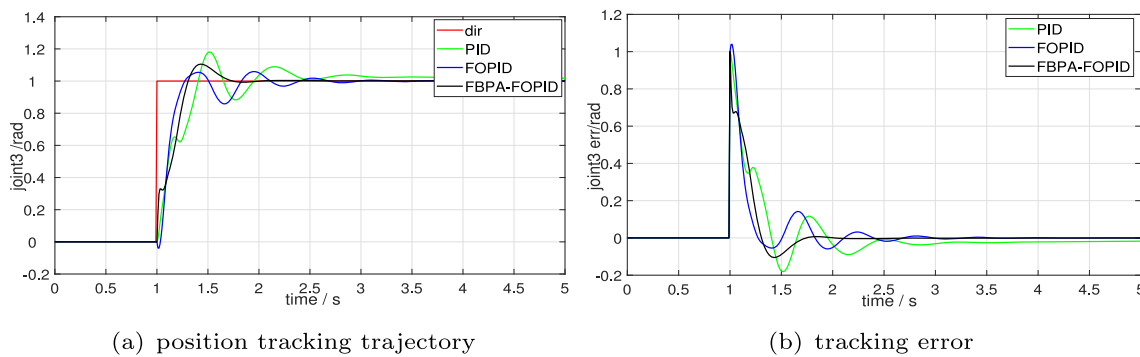
where t represents time, and $e_j(t)$ represents the tracking error of the j -th joint 5.

4 Simulation experiment of robotic arm

In order to verify the control effect of the FOPID controller based on the FBPA proposed in this paper, this paper selects the UR5 6-DOF robotic arm model as the research object and establishes a simulation model of robotic arm trajectory tracking control in the Simulink environment [31]. The inertia and partial physical parameters of the UR5 robotic arm are shown in Table 2. The algorithm parameters in the simulation are set as follows: The beetle population size is 30, the dimension is 5, the maximum number of iterations MaxIter=100, the initial step size step=0.0001, the step size attenuation factor $\eta = 0.95$, the learning factor $c_1 = c_2 = c_3 = 2$, the minimum inertia weight $w_{\min} = 0.4$, the maximum inertia weight $w_{\max} = 0.9$, the minimum particle velocity $v_{\min} = -1$, and the maximum particle velocity $v_{\max} = 1$. The input signal of the control system is a step input. The simulation experiments are carried out using three control schemes: PID control, FOPID control, and FOPID control tuned by the FBPA. The simulation results are compared to observe the tracking effect of the six joints of the robotic arm.

Table 3 Average performance indicators of three controllers with step response

Controller	Overshoot (%)	Adjustment Time (s)	Peak Time (s)
PID	54.6	2.44	1.39
FOPID	31.2	1.89	1.33
FBPA-FOPID	22.1	1.43	1.09

**Fig. 6** Comparison of joint 1 position tracking trajectory and tracking error with step response**Fig. 7** Comparison of joint 2 position tracking trajectory and tracking error with step response**Fig. 8** Comparison of joint 3 position tracking trajectory and tracking error with step response**Table 4** Tracking performance indicators of three controllers with sine response

Controller	Average error of MSE (rad)	Average absolute value input τ (Nm)
PID	2.27×10^{-2}	3.7422×10^4
FOPID	0.88×10^{-2}	2.5686×10^4
FBPA-FOPID	0.37×10^{-2}	2.3154×10^4

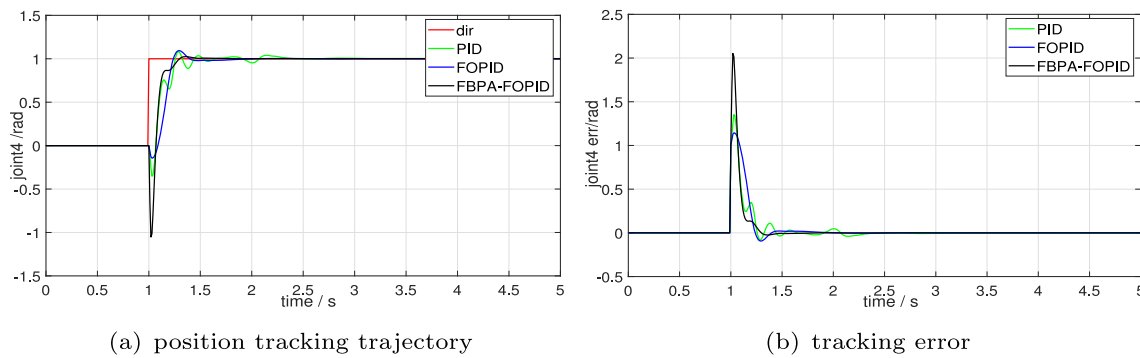


Fig. 9 Comparison of joint 4 position tracking trajectory and tracking error with step response

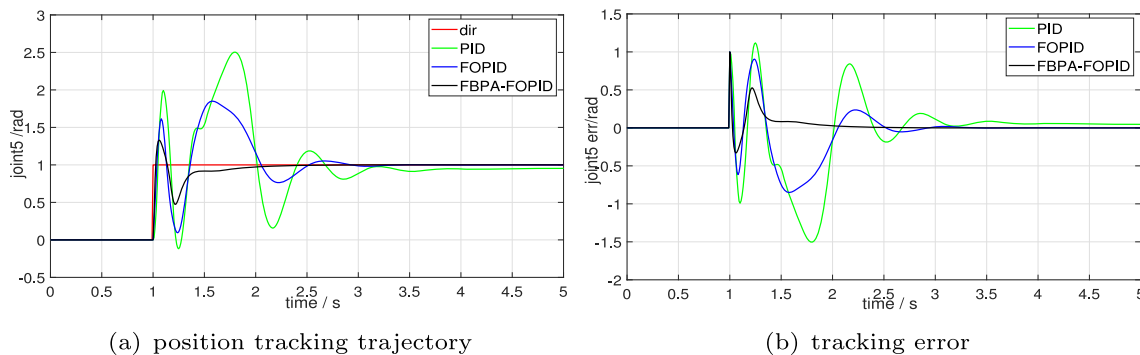


Fig. 10 Comparison of joint 5 position tracking trajectory and tracking error with step response

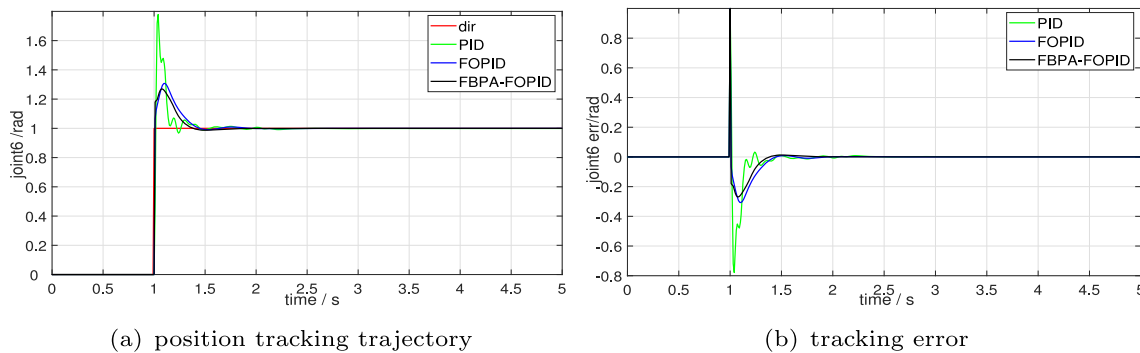


Fig. 11 Comparison of joint 6 position tracking trajectory and tracking error with step response

The tracking results of the six joints are shown in Figs. 6, 7, 8, 9, 10 and 11.

In the simulation experiments, we conducted two sets of experiments: step response and sinusoidal response. The experimental results of the step response are shown in Figs. 6, 7, 8, 9, 10, 11 and 12, while the comparison of the three control methods and the average indicators of the six joints is shown in Table 3; the results of the sine response are shown in Figs. 13, 14, 15, 16, 17, 18 and 19. The tracking performance indicators with the sine response are shown in Table 4.

According to Figs. 6, 7, 8, 9, 10 and 11 and Table 3, the overshoot of more than half of the joints of the integer-order PID control exceeds 30%, and the overshoot of the FOPID

control based on the FBPA is mostly within 30%, which are 28.0%, 20.1%, 10.1%, 2.5%, 31.1%, and 27.5%, respectively. The overshoot is smaller, the adjustment time is shorter, and it is closer to the reference trajectory. On the whole, the tracking error is smaller than that of the other two control methods, the control is accurate, fast and stable, and the trajectory tracking effect is the best. Besides, according to the characteristic curve of joints 1-6 displayed in Fig. 12, the driving torque of the FOPID controller optimized by the FBPA is not too large, and there is no strong chattering during the entire movement of the robotic arm, ensuring the smooth operation of the system.

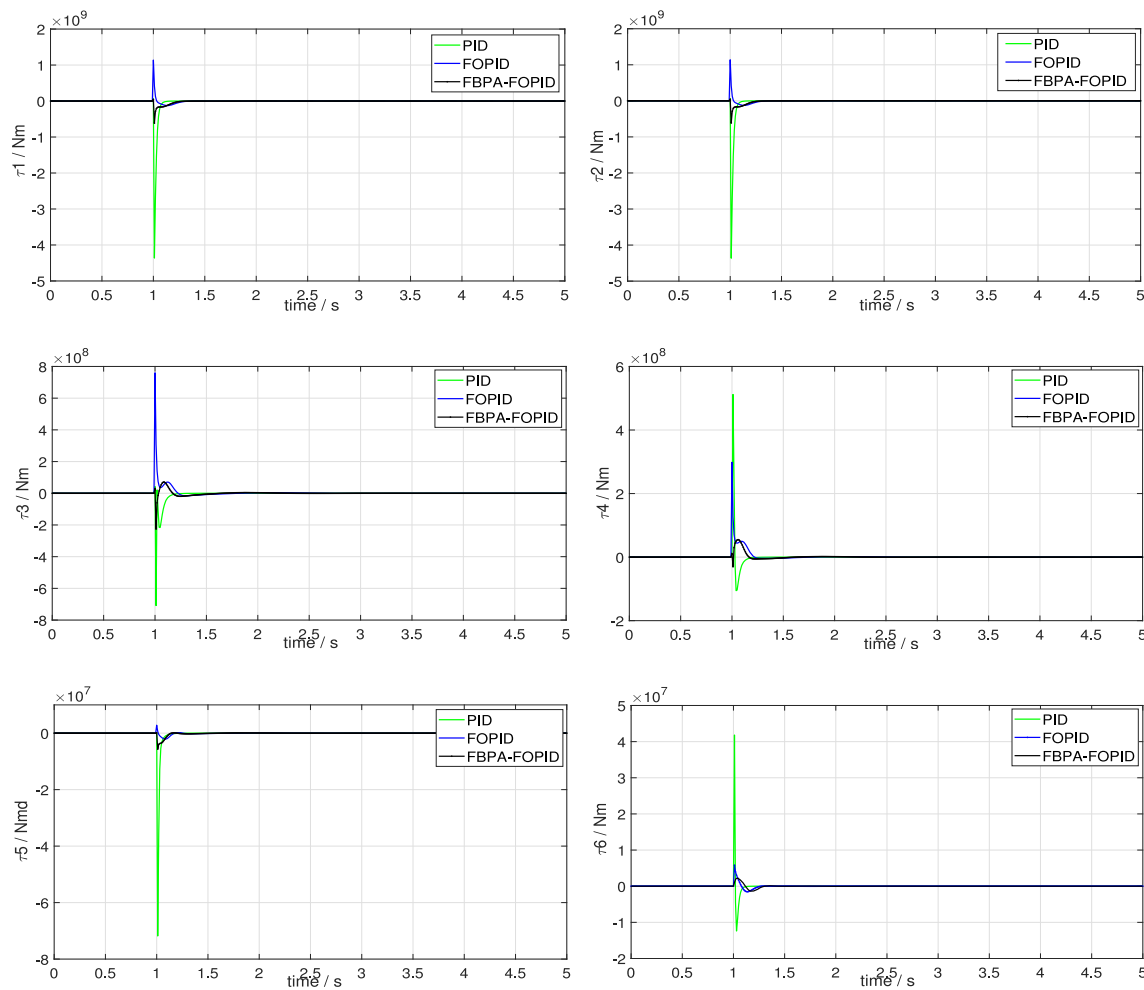


Fig. 12 Joints 1-6 driving torque characteristic curve with step response

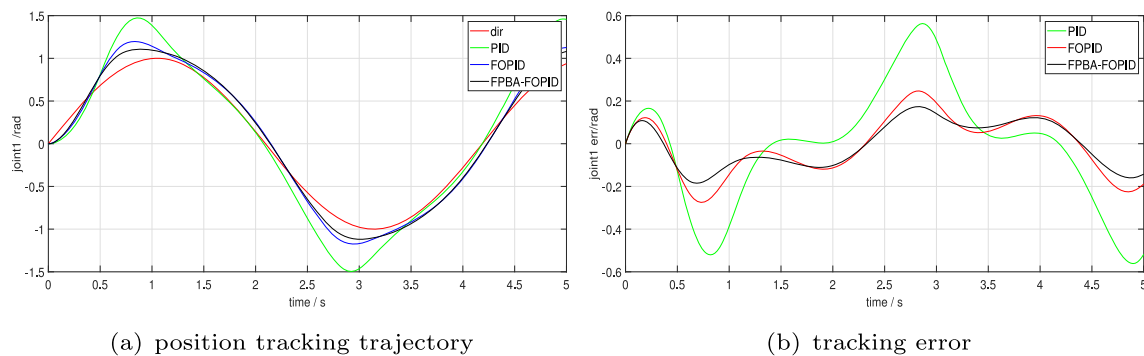


Fig. 13 Comparison of joint 1 position tracking trajectory and tracking error with sine response

As shown in Figs. 13, 14, 15, 16, 17, 18 and 19, the tracking responses of a 6-DOF robotic arm under PID, FOPID and FBPA-FOPID controllers with sinusoidal inputs demonstrate the tracking performance of each joint and the corresponding tracking errors. To evaluate the tracking performance more intuitively, the mean squared error (MSE) of the tracking error for each joint was calculated, followed by averaging

the MSE across all six joints. The results show that the average MSE for the three controllers is 2.27×10^{-2} for PID, 0.88×10^{-2} for FOPID, and 0.37×10^{-2} for FBPA-FOPID. Additionally, to assess the control input efficiency, the absolute values of the input signals for each joint were summed and then averaged across all six joints. The average absolute input values for PID, FOPID, and FBPA-FOPID controllers

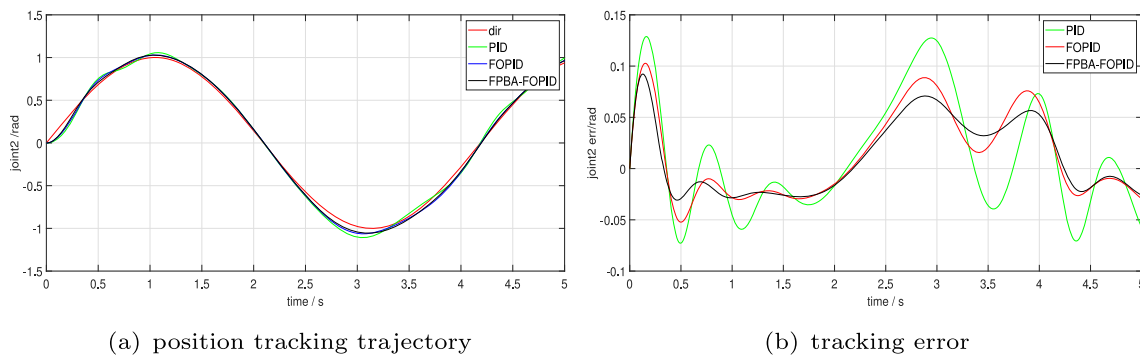


Fig. 14 Comparison of joint 2 position tracking trajectory and tracking error with sine response

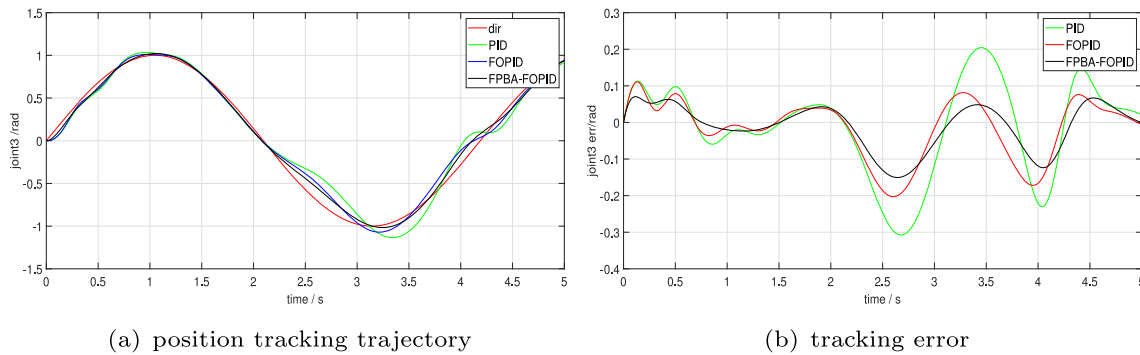


Fig. 15 Comparison of joint 3 position tracking trajectory and tracking error with sine response

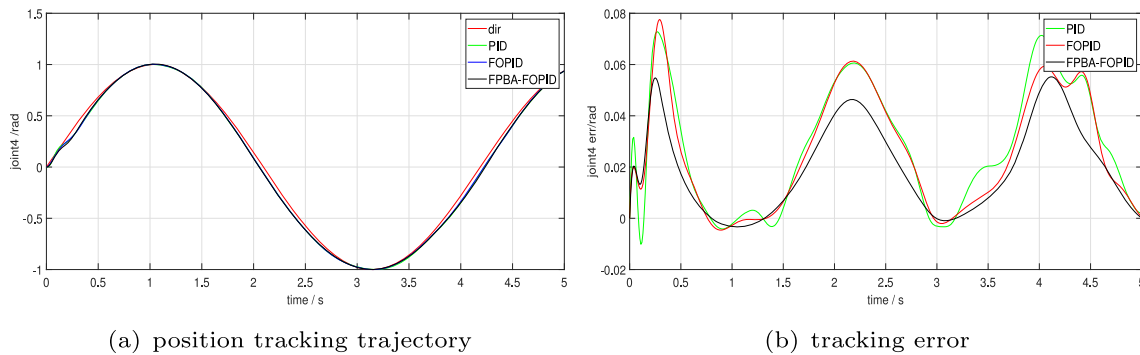


Fig. 16 Comparison of joint 4 position tracking trajectory and tracking error with sine response

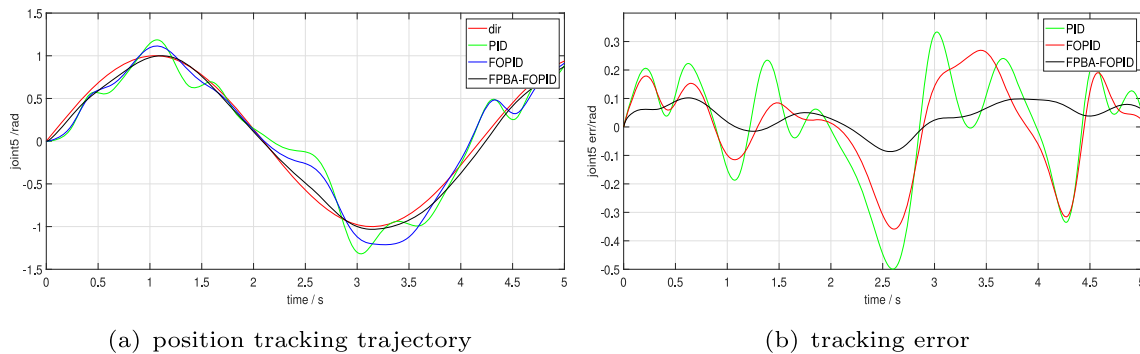


Fig. 17 Comparison of joint 5 position tracking trajectory and tracking error with sine response

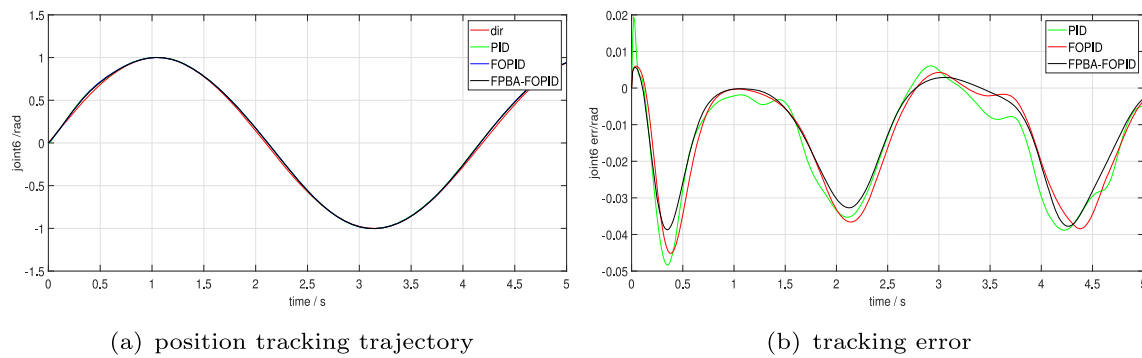


Fig. 18 Comparison of joint 6 position tracking trajectory and tracking error with sine response

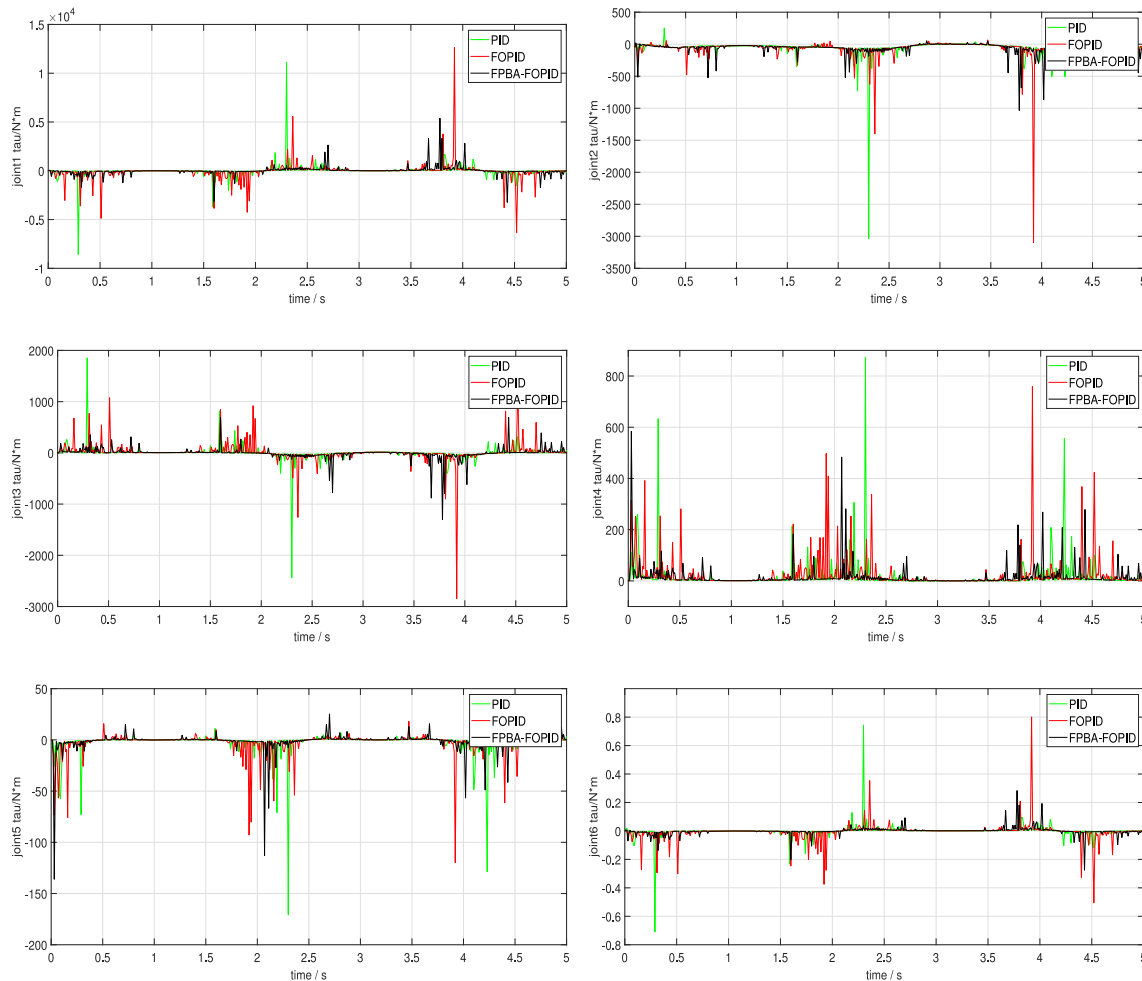


Fig. 19 Joints 1-6 driving torque characteristic curve with sine response

are 3.7422×10^4 , 2.5686×10^4 , and 2.3154×10^4 , respectively. In summary, the FBPA-FOPID controller exhibits superior control performance under sinusoidal inputs, significantly reducing the tracking error while minimizing the control input magnitude, demonstrating higher efficiency and stability.

5 Conclusion

This paper conducts in-depth research and analysis of the dynamics, FOPID controller and trajectory tracking control of the robotic arm, which can be summarized as follows: The dynamic modeling of the robotic arm is the basis for studying the trajectory tracking control of the robotic arm. This

paper takes the 6-DOF robotic arm as the research object, uses the Lagrange method for dynamic analysis and establishes the dynamic model of the 6-DOF robotic arm under the premise of ensuring the dynamic characteristics of the robotic arm. This study explores the basic theory of fractional-order control and its system characteristics and designs a FOPID controller. Through the simulation experiment of trajectory tracking control of the robotic arm model in the Simulink environment, the performance difference between the FOPID and integer-order PID controllers is compared. According to the experimental results, the FOPID controller can describe and control the movement of the robotic arm more accurately. At the same time, in view of the fact that traditional BAS algorithm and PSO algorithm are prone to be trapped into local convergence and premature convergence, this paper introduces the concept of fractional calculus into the algorithm and proposes a fractional-order beetle antennae particle swarm hybrid algorithm, which is applied to the parameter tuning of the FOPID controller and trajectory tracking control of the 6-DOF robotic arm. It greatly reduces the overshoot during the operation of the 6-DOF robotic arm and improves the response speed, obtains higher quality control effect and verifies its effectiveness and reliability in practical applications.

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Author Contributions Zhou Jiang was involved in writing—original draft and methodology and also responsible for formal analysis, supervision, investigation, reviewing and editing. Xiaohua Zhang took part in formal analysis, supervision and visualization. Guoquan Liu participated in formal analysis and visualization. All authors read and approved the final version of the manuscript.

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Availability of data and material The data and code that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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